



Algorithmically designed flaps in tongue reconstruction: a feasibility analysis

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Abstract

Purpose Despite the significance and complexity of tongue reconstruction surgery, a digital tool for flap design is currently lacking. This study investigates the effectiveness of employing inverse finite element method (IFEM) for meticulously designing the geometric characteristics of harvested tissue (free flap) for tongue reconstruction.

Methods In the case of an artificially simulated hemiglossectomy, IFEM algorithm was applied for algorithmic flap design. The method's effectiveness was evaluated by assessing flap deformation in a simplified virtual reconstruction, focusing on parameters such as stress, strain, and thickness.

Results The IFEM algorithm successfully generated an optimal flap design for the intended surgical removal. Analysis of the flap's overall surface area, deformation characteristics, and safety margins demonstrated the feasibility of the deformation. Notably, the stress and thickness assessments suggested that the flap's tension post-surgery would not adversely affect the mobility of the reconstructed tongue, suggesting a positive outcome for functional recovery.

Conclusion The IFEM demonstrates significant potential as a tool for precise free flap design in tongue reconstruction surgeries. Its application could lead to improved surgical accuracy and better quality of life for patients undergoing such procedures.

Keywords Head and neck cancer · Oral cancer · Tongue reconstruction · Precision health · Personalized medicine

Introduction

Oral and pharyngeal cancer constitutes the sixth most common cancer worldwide, posing a substantial and escalating concern [1]. Treating advanced oral cancer can be a challenge, particularly in the case of tongue reconstruction surgery. Surgical resection and reconstruction of the tongue is an accepted procedure to restore the function of the

tongue after cancer removal, yet due to the numerous factors involved, the surgical design of flaps in tongue reconstruction is complicated.

In tongue reconstruction, sufficient volume and tongue surface recovery are reported to be necessary for acceptable outcomes [2]. For instance, treatment protocol design is especially important for the reconstruction of the base of the tongue since it requires a precise bulk recovery [3]. Chepeha et al. [4] highlight that for a successful premaxillary and palatal contact recovery, it is crucial to design the anterior tongue with enough bulk and surface area. This is essential for effective movement and control of food and liquid in the mouth, as well as for clear speech production. Conversely, inefficient bulk and surface recovery can cause disorders and negatively impact functional outcomes. For instance, excessive bulk can cause choking, sleep apnea, and mastication disorders, while insufficient bulk can interrupt the swallowing mechanism and speech production [2]. Moreover, a poorly designed flap could impede the essential

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mobility of the neotongue by tethering the remaining portion of the tongue [5]. Therefore, precise design of flaps is required to ensure acceptable functional outcomes.

The precise design of free flaps results in sufficient bulk and surface area after reconstruction. Currently, there are three commonly used donor sites for tissue transplantation: Anterolateral Thigh (ALT), Rectus Abdominis (RA), and Radial Forearm Free Flap (RFFF) [2]. The ALT and RA donor tissues often provide excessive volume. Although this volume can be trimmed down, the trimming also reduces the skin surface area. This coupling makes precise flap design extremely difficult. Additionally, the muscle of the neotongue might atrophy during the recovery with the ALT and RA donor tissues [6], which makes RFFF an ideal option to achieve the desired functional outcomes [7]. RFFF results in less bulk of tissue, and the remaining bulk required can be tailored by augmenting the fibroadipose tissue of the forearm with the beavertail radial forearm free flap method (BT-RFFF, [2]). This means that one can design the geometry of the flap without worrying about its bulk and then adjust the beavertail fat tissue to tailor the volume of the entire donor tissue.

The BT-RFFF technique allows for the independent design of harvested volume and skin surface area. However, determining the precise geometry of free flaps remains a challenge. Current methods focus on the topology of the defect and attempt to establish a relationship between specific defects and flap shape. Baskin et al. [7] conducted a cadaver study to further investigate this relationship by examining the most common types of tongue reconstruction. They incised the cadaver tongue according to the reconstruction type, resected the region of interest, and debulked it to create an unfolded topology suitable for free flap design. For example, they proposed that a pear-shaped free flap is appropriate for total oral tongue reconstruction, while a bullet-shaped free flap is suitable for oral hemiglossectomy reconstruction. After planning for two surgeries using the predetermined flap shapes, they concluded that these shapes could improve functional outcomes. This study provided valuable insights for further geometrical investigation into the precise design of free flaps.

The literature on patient-specific geometrical flap design in tongue reconstruction is limited, and to our knowledge, only Fan et al. [5, 8] have proposed a method for such a design. In their study, they introduced a novel anatomy-based flap design to improve functional and cosmetic outcomes in subtotal tongue reconstruction. To devise this design, they conducted a virtual geometrical study and collected magnetic resonance imaging (MRI) data from a volunteer's tongue. They identified five landmarks for hemiglossectomy reconstruction and virtually segmented and incised the tongue based on these landmarks. Through geometrical manipulations, they concluded that the optimal shape for a hemiglossectomy free flap is a pentagon. The dimensions of this shape can be determined based on the anatomical

landmarks on the patient's tongue. To evaluate their method, they conducted a cohort study involving thirty-five patients divided into two groups. They designed free flaps for the second group based on the actual anatomical landmarks and compared aesthetic outcomes, swallowing function, and speech intelligibility. They presented their method as a more precise approach to design flaps in hemiglossectomy reconstruction, demonstrating that even simple dimension measurements can enhance outcomes. This suggests that computer-assisted free flap design in tongue reconstruction could lead to more precise surgery. Therefore, further studies are needed to investigate the effectiveness of digital tools for precise flap design.

Surface flattening refers to the process of transforming a three-dimensional (3D) curved surface into a two-dimensional (2D) domain, aiming to preserve the original surface's geometric and topological characteristics as faithfully as possible. This involves geometric considerations, where the inherent properties of the 3D surface, such as shape, size, and connectivity, are retained during the transformation. In computer graphics, surface flattening is utilized for texture mapping, shape analysis, and mesh parameterization [9]. The fabrication industries use it in car body manufacturing [10], furniture making, and clothing (garment) design [11]. It also forms the backbone of flattening-based medical visualizations [12]. Wound reconstruction is the most relevant field where computer-assisted surgical planning has been introduced for precise flap transplantation. To this end, Ji et al. [13] employed a surface flattening technique based on the finite element method (FEM) and utilized the minimization of potential energy to reduce element-wise deformations. Given the successful implementation of surface flattening in similar contexts, it is a promising technique for the objectives of this study.

Although computer-assisted jaw reconstruction has revolutionized surgical practice by reducing operation time, decreasing treatment costs, and improving reconstruction precision [14, 15], there is currently no digital tool available for the algorithmic design of flaps in tongue reconstruction. This makes the surgery heavily dependent on the experience of the surgeons. This study evaluates the feasibility of using a surface flattening algorithm to determine the 2D geometry of the flap required for reconstructing a patient-specific tongue defect. An optimal flap design will enable more accurate surgical planning and increase confidence during the procedure. By standardizing the flap design process, this tool could reduce reliance on individual surgeons and ultimately improve both functional and aesthetic outcomes for patients with head and neck cancer.

Exploring algorithmic strategies for optimal flap design

Central to the surface flattening process is the challenge of minimizing distortion: ensuring that angles, areas, and distances are represented as accurately as possible on the 2D plane. Achieving an optimal balance in this regard is a complex task, often necessitating a strategic compromise between conformal (angle-preserving) and authalic (area-preserving) distortions. It is impossible to fully preserve both area and angle during flattening, so the selection of a technique depends on the specific application and the required qualities. For instance, conformal mapping is preferred in texture mapping to preserve angles and shapes, while optimization-based methods are chosen in fabrication to minimize area distortion and better replicate the physical behavior of materials under deformation.

To achieve an optimal design for free flaps, it is crucial to select a surface flattening technique that minimizes area and angle distortions. These distortions indicate the severity of deformation, and a suitable technique should aim to reduce them to more accurately imitate the real-world behavior of flap deformation. While recent developments in computer graphics, such as boundary first flattening (BFF, [9]), have been effective in preserving area and angles, they do not account for the underlying physics of deformation, surface thickness, and material properties. In physical deformation problems, such as sheet metal forming for car body fabrication, the quality of the final product heavily depends on the initial flat sheet's shape, thickness, and material properties. Therefore, engineers often prefer energy-based algorithms, such as the inverse finite element method (IFEM), which incorporate all these factors. IFEM employs an iterative algorithm to estimate an efficient flat surface, and recent developments have demonstrated its effectiveness in sheet metal forming applications [10]. Nevertheless, it is necessary to evaluate the algorithm's performance specifically in the context of free flap design for tongue reconstruction.

Materials and methods

In the present pre-clinical study, we introduce the IFEM as a promising algorithm for free flap design and investigate its applicability to tongue reconstruction. To demonstrate this, we selected a common oral hemiglossectomy reconstruction as a case study (depicted in Fig. 1), allowing us to showcase the algorithm's potential in a representative scenario. For this study, a standard and well-established tongue model, based on previous modeling works [16–18], was deemed sufficient. Consequently, we adopted this standard model to artificially simulate hemiglossectomy surgical removal and then design an optimal flap for tongue reconstruction. It is

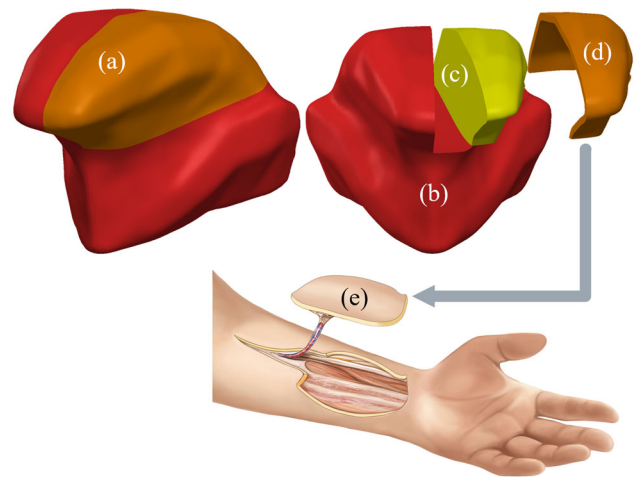


Fig. 1 Schematic representation of resection and free flap in oral hemiglossectomy reconstruction. **a** Indicates the defect requiring surgical removal, while **b** shows the remaining native portion of the tongue. **c** Represents the required beavertail bulk for reconstruction, and **d** shows the outer surface that needs to be recovered with a free flap (**e**)

important to note that this phase of the research involved no real patients and solely utilized the simulated model to assess the algorithm's effectiveness. The model exists in hexahedral discretization form; consequently, our first step was to extract its boundary representation as an STL file (Fig. 2a). Subsequently, we utilized the smoothing and remeshing algorithms available in Autodesk MeshMixer software (www.meshmixer.com) to process this boundary representation and retrieve its organic form. Furthermore, a uniform triangular mesh made the geometry suitable for the flattening algorithms (Fig. 2b).

The next step involves planning the surgical removal and separating this geometry from the rest of the tongue (Fig. 1a). The BT-RFFF can be tailored to fit the specific defect, allowing for a more precise and independent design of free flaps [3]. This flexibility allows for dividing the defect into two subsections, as illustrated in Fig. 1c, d. In this figure, (c) denotes the internal bulk, which is comprised of the fat underneath the skin, the beavertail fat tissue, or a combination of both, while (d) represents the outer surface, which is reconstructed using the skin of the free flap (e). This surface is subsequently re-meshed with uniform membrane triangle elements using Autodesk MeshMixer software (Fig. 2d). The surface is then introduced to the BFF algorithm to generate a reasonable initial guess (Fig. 2e), serving as an input for the IFEM algorithm.

Figure 3 illustrates the fundamental concept of the IFEM algorithm. In this inverse approach, we start with the known final deformed state of the flap as a 3D surface (Fig. 1d). Additionally, the thickness of the undeformed free flap is assumed to be known; for this study, we adopted a thickness of two millimeters for the flap as suggested by Baskin et al.

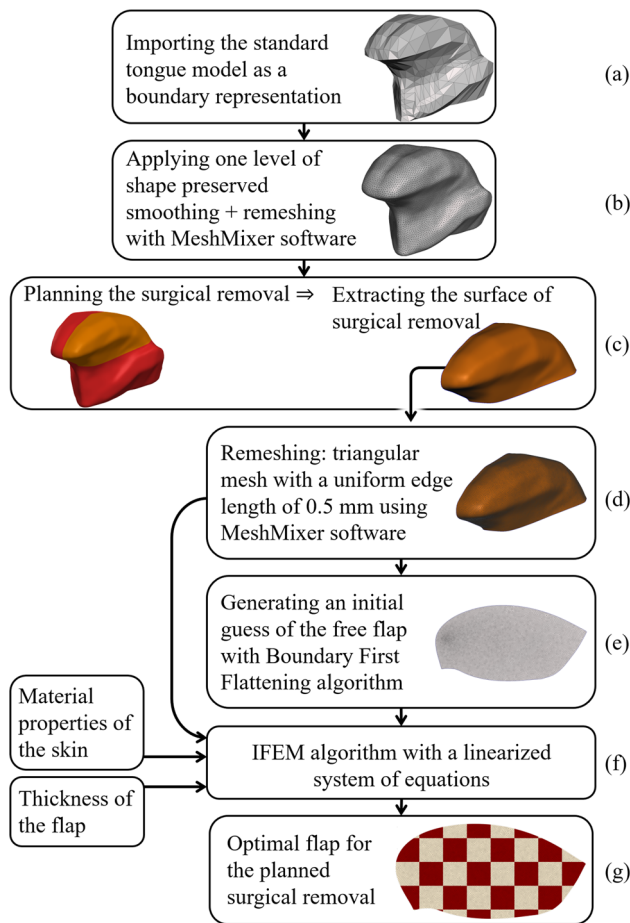


Fig. 2 The overall workflow: from geometry pre-processing to optimal flap design using the IFEM algorithm

[7]. The objective is to estimate an optimal flap that can be deformed into the final 3D shape while minimizing deformation energy. The specific IFEM algorithm utilized in our study is adapted from Bostan Shirin et al. [10] with a linearized system of equations. Such linearization enhances the algorithm's robustness and ensures convergence, without compromising on accuracy [19]. This approach implements linearization in each iteration, adeptly capturing the material's nonlinear behavior through the iterative inverse process.

The IFEM algorithm employed in our study is formulated based on the principle of minimum potential energy and operates under several key assumptions. Firstly, it conceptualizes the deformation as a plane stress problem, focusing solely on the initial and final states of the deformation without considering intermediate stages. This approach is iterative: At the outset, the undeformed body is taken as the initial guess, and through successive iterations, this model is refined until it converges to the optimal flap design. One of the algorithm's significant strengths is its capacity to calculate strains, stresses, and thickness distribution within the deformed flap. This capability is instrumental in evaluating the severity of

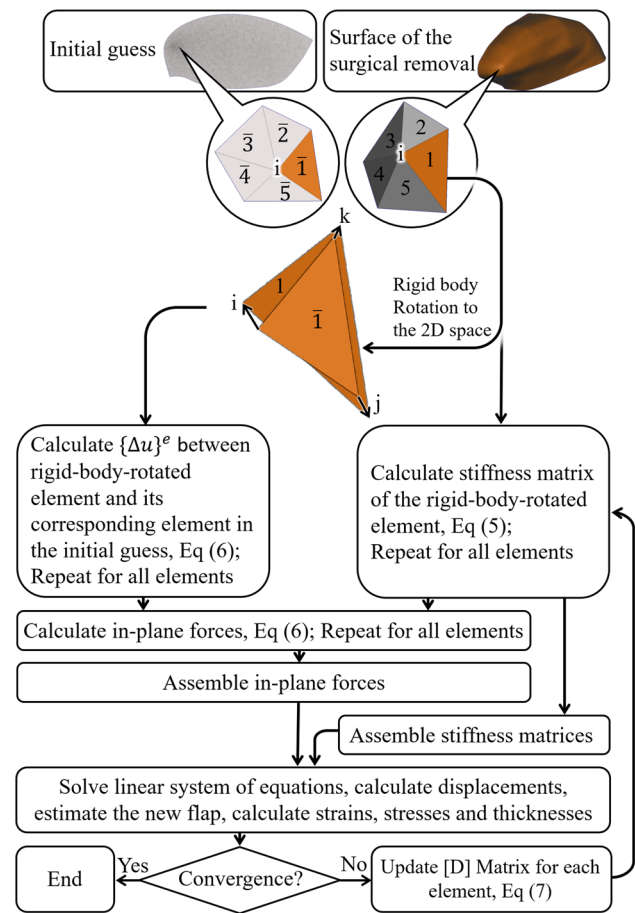


Fig. 3 Flow diagram of IFEM algorithm used in this study

deformation and ensuring that the designed flap closely mimics the physical behavior of natural tissues. However, it is important to note a critical limitation: The IFEM algorithm in its current form is exclusively focused on the deformation of the free flap. It does not account for the deformation of other components of the tongue, such as the muscles and the beavertail fat tissue.

What follows is a brief explanation of the kinematics and finite element formulation of the IFEM algorithm. In the final configuration, the increment of potential energy is formulated as follows [10]:

$$\delta\psi = \sum_{e=1}^n \int_{V^e} \{\delta\varepsilon^e\}^T \{\sigma^e\} dV - \sum_{e=1}^n \int_{S^e} \{\delta u^e\}^T \{T^e\} dS \quad (1)$$

where $\{\delta u\}$, $\{\sigma\}$, $\{\varepsilon\}$, and $\{T\}$ represent the incremental displacement of the nodes, the stress vector, the strain vector, and the boundary tractions, respectively.

The relationship between the stress vector and the displacement vector is expressed as follows:

$$\{\sigma^e\} = [D^e]\{\varepsilon^e\} = [D^e][B^e]\{u^e\} \quad (2)$$

where $[D]$ represents the material property matrix, and $[B]$ is the strain operator, as defined below for a membrane triangular element in the two-dimensional space, with y_{jk} representing the difference $y_k - y_j$. The indices i, j, k denote the local nodal numbering on the triangular element.

$$[B^e] = \begin{bmatrix} -y_{jk} & 0 & -y_{ki} & 0 & -y_{ij} & 0 \\ 0 & x_{jk} & 0 & x_{ki} & 0 & x_{ij} \\ x_{jk} & -y_{jk} & x_{ki} & -y_{ki} & x_{ij} & -y_{ij} \end{bmatrix} \quad (3)$$

As stated, $[B]$ is considered linear in each iteration. However, ultimately, the IFEM captures the nonlinearity of the material behavior through an iterative inverse approach. Based on the principle of virtual work and considering triangular membrane elements, Eq. (1) yields the following [10]:

$$\sum_{e=1}^n [B^e]^T [D^e] [B^e] \{u^e\} A^e t^e - \sum_{e=1}^n \{F^e\} = 0 \quad (4)$$

in which $\{F\}$ are the external forces, calculated in each iteration based on the updated material property matrix, area A^e , and thickness t^e of each element. In every iteration, nodal displacements are determined by solving this linear system of equations.

Figure 3 briefly describes the iterative approach in the IFEM algorithm. It first takes the 3D surface and an initial guess for the solution as two inputs. Since equilibrium should be satisfied in the deformed body, all elements of the 3D surface are transformed to the 2D space through a rigid body rotation and are subsequently considered for stiffness matrix calculations as below:

$$[k^e] = [B^e]^T [D^e] [B^e] A^e t^e \quad (5)$$

External forces $\{F\}$ are then assumed to be planar and are generated during the flattening of the deformed body. The calculation of the external forces requires determining the in-plane forces for each element:

$$\begin{aligned} \{f_{\text{in-plane}}^e\} &= [k^e] \{\Delta u^e\}, \quad \{\Delta u^e\} \\ &= \{x_{\text{initial guess}}\}^e - \{x_{\text{transferred}}\}^e \end{aligned} \quad (6)$$

in which $\{x_{\text{initial guess}}\}^e$ are the nodal coordinates of the transferred element, and $\{x_{\text{initial guess}}\}^e$ denotes the nodal coordinates of the corresponding element in the initial guess. Having determined the in-plane forces for all elements, the

external force at each node is calculated by summing the contributions from all elements sharing that node. For instance, in Fig. 3, the external force at node “i” is calculated as the resultant of adding the contributing in-plane forces from elements 1 to 5 at node “i.” Therefore, in Eq. (4) the only unknowns are the material properties matrix $[D]$ and the thickness (t^e) for each element. Initially, the thickness of all elements is set equal to the thickness of the free flap. This thickness is updated at the end of each iteration. Moreover, the material properties matrix is calculated for a plane stress problem as follows:

$$[D]^e = \frac{\hat{E}^e}{1 - v^2} \begin{bmatrix} 1 & v & 0 \\ v & 1 & 0 \\ 0 & 0 & \frac{1-v}{2} \end{bmatrix} \quad (7)$$

where $\hat{E}^e = f(\bar{\sigma}, \bar{\varepsilon}) = \frac{\bar{\sigma}}{\bar{\varepsilon}}$ is the corrected Young's modulus used to capture each element's nonlinear behavior through iterations. Here, $\bar{\varepsilon}$, $\bar{\sigma}$, and v represent the equivalent strain, equivalent stress, and Poisson ratio of the skin, respectively. For the incompressible hyperelastic skin, $v = 0.5$ is assumed, and the equivalent strains and stresses are updated at the end of each iteration. For the initial step, $\hat{E}^e = 1$ Mpa is selected for all elements. This arbitrary initial value does not affect the solution, and the iterative approach converges to its solution regardless of this value [10].

Assembling the in-plane forces and stiffness matrices allows us to solve the linear system of Eq. (4) and calculate the nodal displacements. The initial guess of the flap is then updated, resulting in a new estimation. Given the initial and final configurations of the deformation, we measure its severity through element-wise calculation of stretches using the continuum mechanics of large deformations, as explained in [20]. This process yields the equivalent strain $\bar{\varepsilon}$ for each element, which is then used to calculate the equivalent stress $\bar{\sigma}$ based on the actual material behavior of the skin. We adopted this relationship from an experimental cadaver study by Annaidh et al. [21], considering the average of all measurements and approximating the stress–strain relationship with a power function ($\sigma = K\varepsilon^n$, where σ , ε , K , and n are true stress, true strain, first, and second constant, respectively). For this feasibility study, we assumed that the anisotropic behavior of the forearm skin is not predominant and only considered the isotropic behavior. The constants were approximated to values of 140.9 MPa and 2.6, with the coefficient of determination (R^2) for this approximation being 99.14%. They also reported a mean failure true strain (0.43) and mean ultimate tensile strength (21.6 MPa), which we considered as thresholds for safety factor calculation.

The calculated $\bar{\varepsilon}$ and $\bar{\sigma}$ are then used to update each element's material properties matrix (Eq. 7), the thickness distribution on the final body, and the stiffness matrix of all

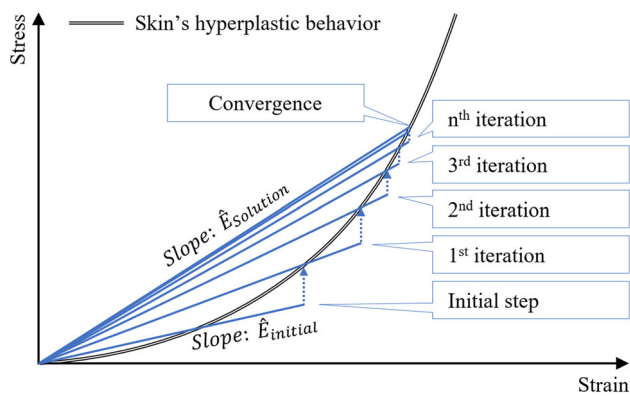


Fig. 4 Iterative convergence process for the corrected Young's modulus

elements (Eq. 5). The iterative process recalculates the in-plane forces and solves the new linear system of equations to estimate a new flap design (see Fig. 3). Convergence is achieved when for all elements there is a negligible change in the corrected Young's modulus between two subsequent iterations:

$$\frac{\hat{E}_{\text{iteration}(n)} - \hat{E}_{\text{iteration}(n-1)}}{\hat{E}_{\text{iteration}(n-1)}} < 0.001 \quad (8)$$

Figure 4 demonstrates this iterative process up to the convergence point for one element. Initially, the value of $\hat{E}^e$ results in a calculated value for $\bar{\epsilon}$, which is then used to find the corresponding $\bar{\sigma}$ based on the actual material behavior. These two values are used to calculate the slope for the next iteration ($\hat{E}^e = \frac{\bar{\sigma}}{\bar{\epsilon}}$).

IFEM is an iterative algorithm that employs the principle of minimizing potential energy to estimate the optimal undeformed configuration. Consequently, the converged step of the IFEM procedure yields an optimal flap design. In practical terms, this means that the deformation of the designed flap (Fig. 1e) to the curved surface (Fig. 1d) results in minimal deformation energy, closely mimicking the skin's physical behavior.

To assess the effectiveness of the designed flap, we investigated the strain, stress, and thickness distribution on the deformed body. These calculated outcome variables are indicative of the deformation's severity and provide insights into the potential tethering of the tongue post-surgery. The inclusion of thickness distribution as a criterion is crucial because thinning of the flap signals bi-axial stretches in the deformed body. These stretches can lead to tensions, potentially resulting in the tethering of the tongue, thereby reducing mobility, and causing functional disorders. For example, stretching a flap in two perpendicular directions can lead to it becoming thinner, generating stresses, and ultimately resisting further stretching. Such resistance might cause tethering in the new tongue. Additionally, local thickening of the flap

could lead to wrinkles, which are aesthetically undesirable and may cause accidental injuries due to occlusion of the jaw. Therefore, post-reconstruction, it is desirable to achieve minimal stress and an even thickness distribution across the flap to mitigate any unwanted tensions. The visual and statistical representations of these field variables are detailed in the Results section.

Results

In this study, we focused on a hemiglossectomy reconstruction as a representative case study. Initially, we artificially simulated a hemiglossectomy surgical removal, followed by designing an optimal flap for tongue reconstruction. To accurately define the defect, two incisions were made in the tongue model. Figure 5b displays the coronal view of the tongue, illustrating a planar cut intersecting with the sagittal plane. The second incision, which follows a curved path, is depicted as a curved line in the sagittal view (Fig. 5c), effectively separating the defect from the rest of the tongue. We discretized the surface of this defect using triangular membrane elements, resulting in a model with 10,532 nodes and 20,698 elements.

Employing the IFEM algorithm, we generated an optimal free flap for the planned surgical removal. The computational process was executed on a standard desktop computer, taking approximately 90 s to reach a converged state after 8 iterations. The designed flap's surface area closely matched that of the resection area (0.7% difference). The outcome of this algorithmic design process is presented in Fig. 5d, including isometric, coronal, and sagittal views of the neotongue in Fig. 5a–c. For illustrative purposes, we applied a one-to-one mapping of a chess pattern to demonstrate the transformation from the 2D flap to the surgical removal. Additionally, the inclusion of landmarks (I), (II), and (III) across different views aids in orienting the reader and correlating various regions of the flap with the surgical removal.

Figures 6, 7, and 8 illustrate the results of the deformation from the 2D flap to the spatially curved surface. Figure 6a shows the color-coded thickness distribution of the deformed flap on the neotongue. Accompanying this, Fig. 6c, d displays the histogram and statistical box plot of the thickness distribution, respectively. In these plots, the dashed line indicates the average, while the solid line represents the median of the distribution. Figures 7 and 8 follow a similar presentation format, showcasing the equivalent strain and stress distributions, respectively. Since the heatmaps on the 3D geometry present some blind spots, we have adapted these heatmaps for 2D representation, effectively mapping them onto the 2D flap layout. This modification not only illuminates the previously obscured areas but also provides a clearer and more comprehensive view of which regions on

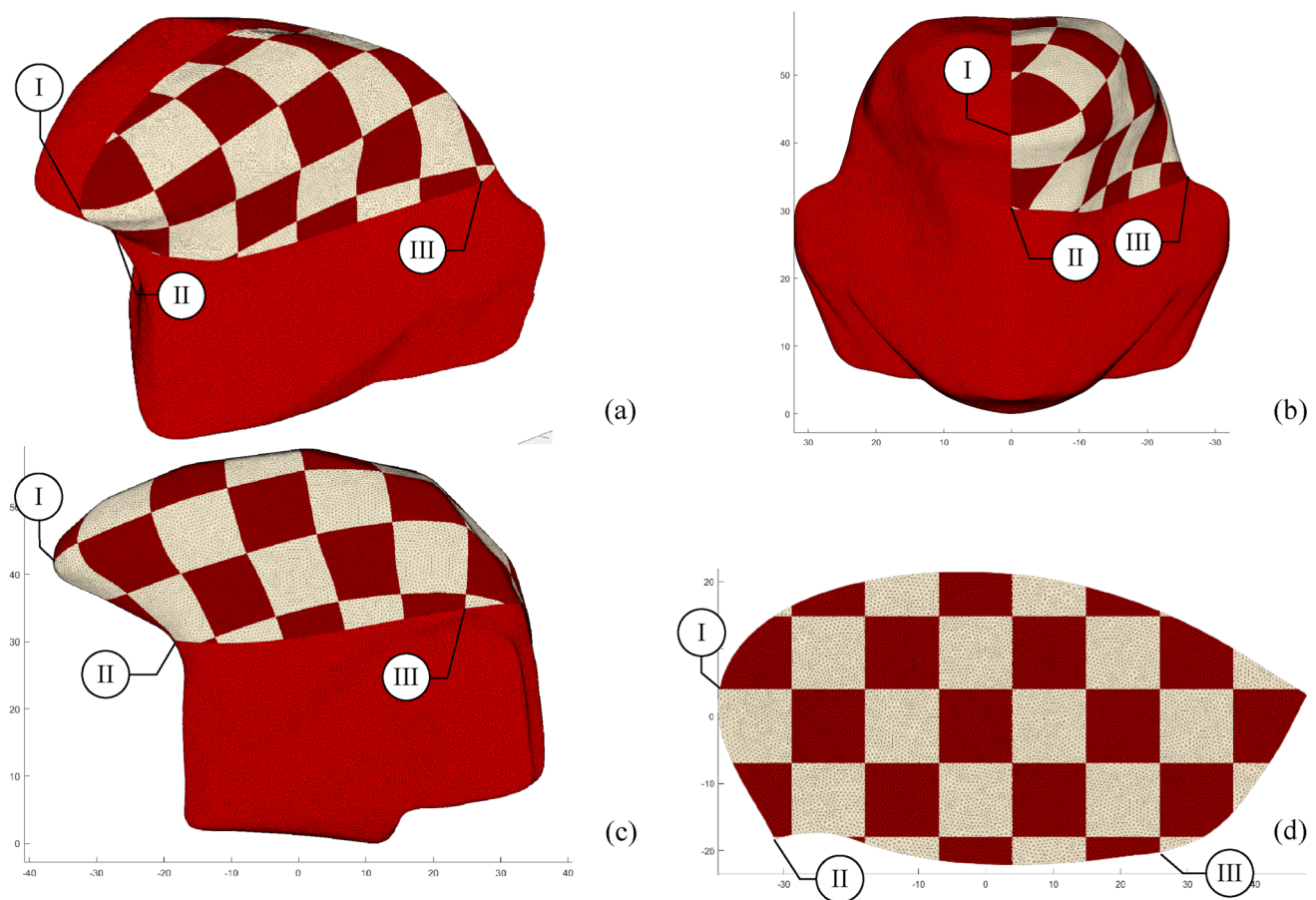


Fig. 5 Depiction of the artificially simulated hemiglossectomy defect and the corresponding free flap design in this case study. **a** Provides an isometric view of the tongue with the defect displayed in different color. **b**, **c** Display the same geometry from coronal and sagittal views, respectively. **d** Illustrates the designed flap using IFEM algorithm. (I)

indicates the anterior mid-sagittal margin corresponding to the tip of the tongue. (II) indicates the anterior–inferior margin on the mid-sagittal plane. (III) indicates the posterior–lateral margin of the resection. All dimensions are in millimeter

the flap are likely to undergo the most significant deformation post-reconstruction. All these plots were generated using MATLAB (MathWorks, Inc., Natick, MA, USA). Additionally, Table 1 provides a summary of the mean, median, and standard deviation of these distributions.

Discussion

The importance of accurately designing the shape and dimensions of the flap is paramount, as imprecise designs can lead to undesirable aesthetic outcomes and diminished functionality, particularly due to the tethering of the tongue. In this work, the success of the flap design was evaluated by measuring the thickness, stress, and strain distribution post-reconstruction. Ideally, after reconstruction, all stress and strain values would be zero, and the flap would maintain its initial thickness.

However, achieving such a scenario is practically unattainable when reconstructing a 3D curved surface using a flat flap. A certain degree of deformation is inevitable in order to mold the flat flap to the complex topography of the 3D surface, which naturally leads to some compromise in stress and strain distribution. The novel method of this work is designed to minimize these compromises. We compared the observed results with both the theoretical ideal and the known thresholds of skin tissue behavior. Equally crucial is the flap's overall surface area, which should closely mirror that of the surgical removal. This agreement is vital for facilitating the recovery of essential interactions with other oral structures. In assessing the design's effectiveness, we took into account all these outcomes, striking a balance between the ideal and the practical.

One of the key achievements of this study is the demonstrated capability of the IFEM in designing a free flap for the planned surgical removal. The chess pattern on the flap

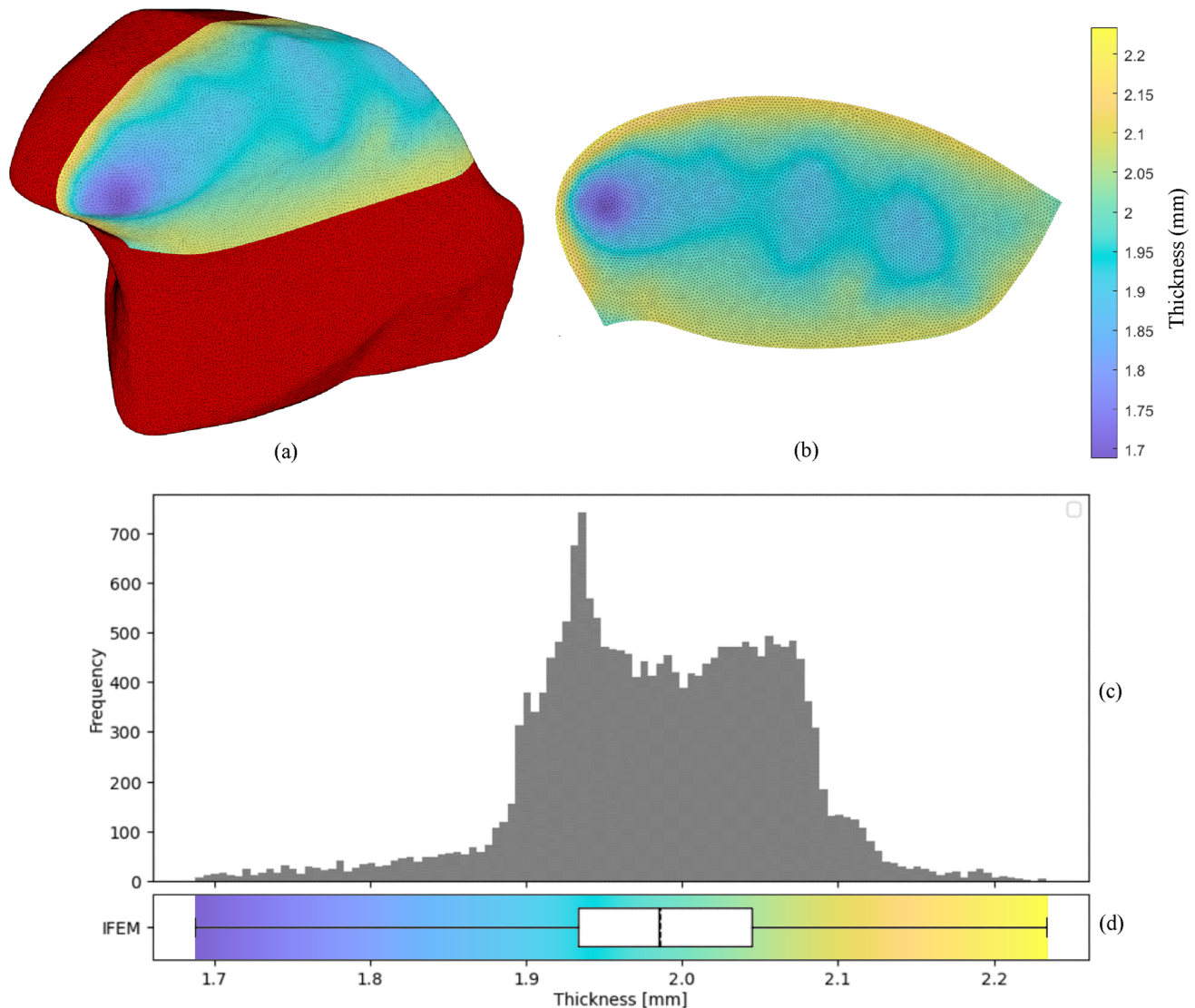


Fig. 6 Thickness distribution of the designed flap after deformation. **a** Illustrates the color-coded thickness distribution, **b** depicts this distribution mapped onto the 2D flap, **c** presents a histogram of the

distribution, and **d** shows a statistical boxplot of the distribution. The reader is referred to the web version of this paper for the color representation of this figure

(Fig. 5) aids in visualizing the flap's deformation after reconstruction. This visual aid suggests that the flap's deformation aligns well with our intuitive understanding of it. Furthermore, the measurement of the flap's overall surface area supports the rationale that the deformation is reasonable. The surface area of the designed flap (2964 mm^2) closely matches that of the surgical removal (2943 mm^2). Such a close match is a strong indicator that the deformation of the designed flap into its final shape is consistent with the incompressibility properties of skin tissue. This finding supports the feasibility of the flap conforming to the curved surface of the resection area without significant compression or expansion, which is a critical consideration in the success of such reconstructive procedures. It is worth noting that the maximum deviation

of the flap designed using IFEM from the initial guess is only 3 mm, suggesting that even the boundary first flattening (BFF) algorithm surpasses mere guesswork in flap design. However, the most precise flap design is achievable with IFEM. While these results appear acceptable, we conducted further investigations to quantitatively assess the quality of this deformation.

The success of the flap design can be gauged by measuring the thickness distribution after reconstruction, which is the most tangible metric. The ideal scenario is where the flap's thickness remains consistently close to 2 mm—the initial thickness—showcasing a narrow distribution and minimal deviation. Such uniformity is indicative of minimal bi-axial stretching, which is crucial as it suggests a reduced likelihood

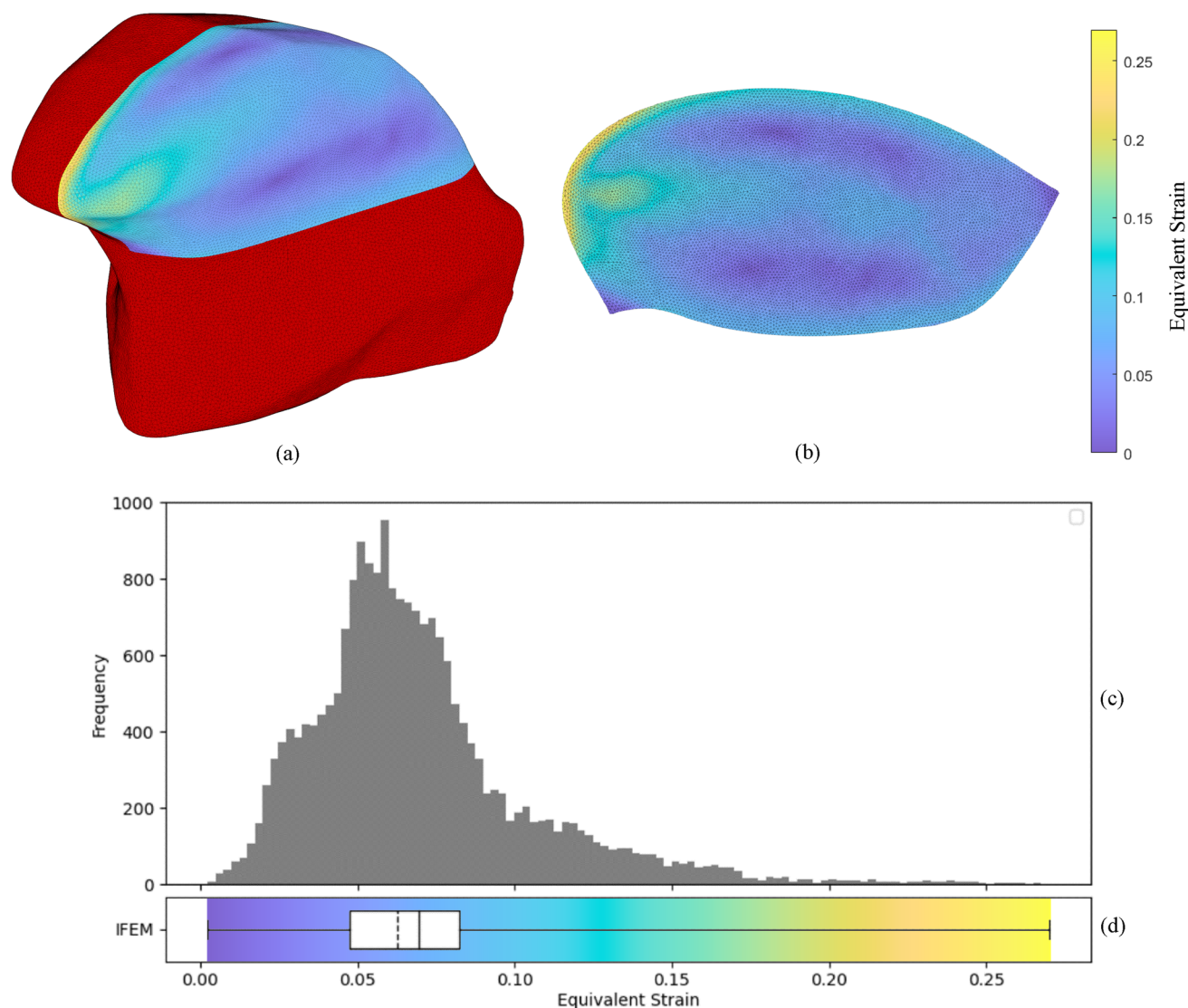


Fig. 7 Equivalent strain distribution of the designed flap after deformation. **a** Illustrates the color-coded equivalent strain distribution, **b** depicts this distribution mapped onto the 2D flap, **c** presents a histogram of the

distribution, and **d** shows a statistical boxplot of the distribution. The reader is referred to the web version of this paper for the color representation of this figure

of post-surgical tethering of the tongue. The histogram in Fig. 6b shows a concentration of thickness values around two millimeters with a small standard deviation, as detailed in Table 1. A balanced distribution without significant skewness is key. Should the histogram show a skewness to the left or right, it would indicate excessive thinning or thickening of the flap post-deformation. This is a critical point as it implies an inadequate design, potentially leading to tethering and wrinkling in the reconstructed tongue. In essence, an ideal thickness distribution is one that closely mirrors the original thickness of the flap, ensuring both functional efficiency and aesthetic integrity in the postoperative outcomes.

The thinnest area appears in the region of high curvature, specifically at the anterior–lateral tip of the reconstructed

tongue, where it thins down to 1.7 mm. Several factors could contribute to this thinning. First, our simulation only accounted for the deformation of the flap. A significant portion of the thinning could be counterbalanced by including the deformation of the native tongue and the fat tissue in a more comprehensive simulation. Secondly, the material properties used in this research were adopted from a study using cadavers that were, on average, 4 days old and from a very elderly population. This suggests that the skin material properties in our study might be stiffer than those of fresh skin. Incorporating more accurate skin properties could potentially address this thinning issue. It is worth mentioning that the thickness of the majority of elements is very close to the initial thickness of the flap, with 85% of elements falling

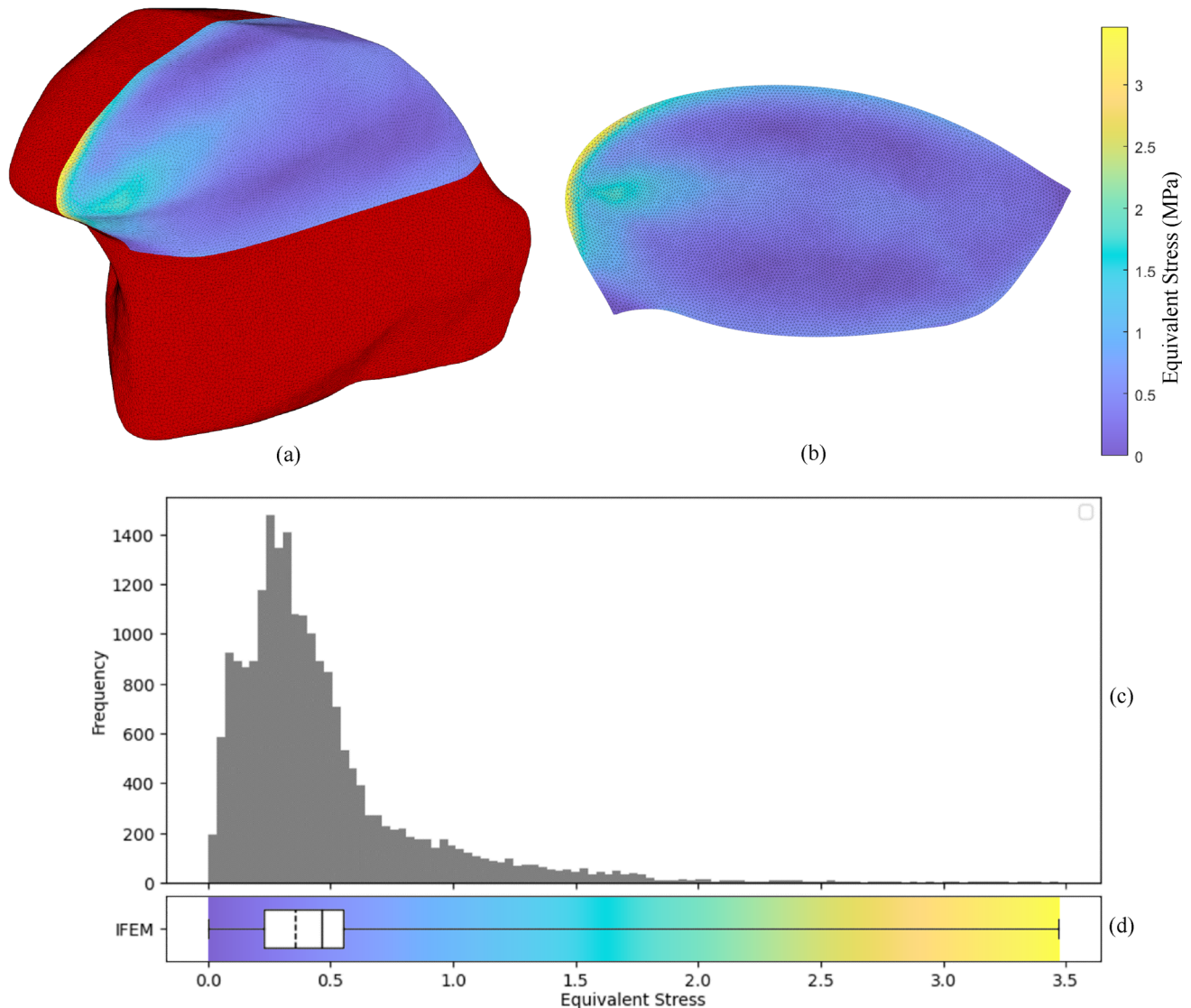


Fig. 8 Equivalent stress distribution of the designed flap after deformation. **a** Illustrates the color-coded equivalent stress distribution, **b** depicts this distribution mapped onto the 2D flap, **c** presents a histogram of the

distribution, and **d** shows a statistical boxplot of the distribution. The reader is referred to the web version of this paper for the color representation of this figure

within a range of 1.9 to 2.1 mm. This indicates that the overall deformation is within acceptable limits.

Equivalent strains and stresses serve as vital indicators of the severity of deformation. Lower values are preferred to prevent tissue damage and undesired tensions in the neotongue, as they may result in reduced mobility. According to the histogram of the equivalent strain distribution in Fig. 7c, the maximum strain (0.27) is significantly less than the threshold (0.43), indicating that the overall deformation is within safe limits. Equivalent stress is a more reliable predictor of tissue rupture. As shown in Fig. 8a, high stresses occur at the tip of the tongue on the mid-sagittal plane. These stresses are compressive and occur due to the high curvature

of the deformed flap in that area. In other words, the flap's border is shaped into a high curvature, resulting in compression. The thickening of the flap in that area also confirms this observation. It should be noted that in a more comprehensive simulation, which accounts for factors such as the deformation of the beavertail fat tissue and the native portion of the tongue, most of these compressive stresses would likely be alleviated. However, this hypothesis warrants further investigation in future studies.

Maximum tensile stresses do not exceed 2 MPa, which is ten times lower than the threshold for damaging the tissue. This safety factor of ten assures that the deformation is safe. Additionally, these relatively low stresses imply that

Table 1 Mean, median, and standard deviation of results

Descriptive statistics of the distribution	Value
Thickness (mm)	
Mean	1.985
Median	1.986
Standard deviation	0.006
Equivalent strain	
Mean	0.069
Median	0.063
Standard deviation	0.0012
Equivalent stress (MPa)	
Mean	0.464
Median	0.357
Standard deviation	0.156

only minor deformations are anticipated when the neotongue is in a relaxed state. This means that, upon completion of the reconstruction, the relatively low stresses would result in minimal deviations from the intended final shape of the neotongue, thereby enhancing its aesthetics and mobility. It is vital to reiterate that including the native tongue's deformation could significantly reduce extreme conditions and further alleviate strains and stresses. Moreover, using material properties that accurately reflect fresh skin would simulate a less stiff skin model, leading to reduced stress levels and improved mobility in the reconstructed tongue. Considering these factors, we can deduce that the simplified simulation used in this study provides the most conservative, yet effective, evaluation of flap deformation in a virtual reconstruction simulation.

We acknowledge several limitations to this study, which could be seen as opportunities for future research. Firstly, we relied solely on a standard, well-established tongue model from the literature, the anatomical accuracy of which leaves room for improvement. Ideally, we would collect multiple images from patients undergoing tongue reconstruction surgery. Secondly, the mid-sagittal glossectomy resection line typically extends inferiorly to the floor of the mouth. Though this surgical perspective is important for a real hemiglossectomy reconstruction, for the purposes of this feasibility study, our example is sufficient. Additionally, we examined flap design only for hemiglossectomy; future studies should consider other types of tongue reconstruction. In our virtual reconstruction, the only deformable continuum was the flap. A more comprehensive simulation could model the deformation of the remainder of the neotongue and investigate the resting state of the neotongue. While our virtual reconstruction examined three measurements and

interpreted the results in terms of their impact on the quality of the neotongue, additional measurements—such as the relative values of the principal strains and their potential to predict rupture and wrinkles during reconstructions—may also be beneficial. Further objectives for future research could include considering the nonhomogeneity and directional properties of the flap, measuring the stiffness of the reconstructed neotongue, conducting mobility analysis of the neotongue, examining more tongue postures, and carrying out experimental validations.

Conclusion

The choice of IFEM for achieving an acceptable flap design has shown effective, as the algorithm precisely designed a free flap while preserving the required surface area for reconstruction. The results of the virtual reconstruction confirmed this, as the thickness, strain, and stress analyses demonstrated acceptable flap deformations. The safety factor of ten implies that the flap designed is safe for use in reconstruction, and the conservative overestimation of strains and stresses in the simplified simulation adds a layer of assurance. IFEM, taking into consideration the thickness and material properties of the flap along with the underlying physics of deformation, proves to be an apt digital tool for accomplishing optimal flap design in tongue reconstruction. The development of digital tools for precise flap design has the potential to enhance the precision and efficiency of tongue reconstruction, possibly leading to improved functional outcomes.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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